Changing Word Equations to Symbol Equations

Copper + oxygen gas → Copper(II) oxide



Iron + Hydrochloric acid → Iron (III) chloride

Silver + Hydrochloric acid → silver chloride

Zinc + oxygen gas → zinc chloride

Sodium + chlorine gas → sodium chloride

Calcium + chlorine gas → calcium chloride

Carbon + zinc oxide → carbon dioxide + Zinc metal

Methane + oxygen → carbon dioxide + water

silver oxide + potassium → potassium oxide + silver metal

chromium + chlorine → Chromium(III) chloride

carbon + lead oxide → carbon dioxide + lead metal

nitrogen gas + Potassium → Potassium nitride

Data Table

1. Metals. For metals we just write the symbol on its own. So iron is written as Fe.
2. There are some formula that we need to know.

	Name	Formula
The diatomic gasses	Oxygen gas	O_2
	Nitrogen gas	N_2
	Hydrogen gas	H_2
	Chlorine gas	Cl_2
	Carbon dioxide	CO_2
	Water	H_2O
	Carbon	C

3. ACIDS

Hydrochloric Acid	HCl	Sulfuric Acid	H_2SO_4
Nitric Acid	HNO_3	Phosphorous Acid	H_3PO_3

IONS

CATIONS. THE POSITIVE IONS

1+ Charge		2+ Charge		3+ Charge	
H^{1+}	hydrogen	Mg^{2+}	magnesium	Al^{3+}	aluminum
Na^{1+}	sodium	Ca^{2+}	calcium	Fe^{3+}	iron(III)
K^{1+}	potassium	Ba^{2+}	barium	Cr^{3+}	chromium(III)*
Cu^{1+}	copper(I)	Zn^{2+}	zinc		
Li^{1+}	lithium	Cu^{2+}	copper(II)		
Ag	silver	Pb^{2+}	lead(II)		
		Fe^{2+}	iron(II)		

ANIONS THE NEAGATIVE IONS

1– Charge		2– Charge		3– Charge	
F^{1-}	fluoride	O^{2-}	oxide	PO_4^{3-}	phosphate
Cl^{1-}	chloride	S^{2-}	sulfide	P^{3-}	phosphide
Br^{1-}	bromide	SO_4^{2-}	sulfate	N^{3-}	nitride
I^{1-}	iodide	SO_3^{2-}	sulfite		

NO_3^{1-}	nitrate	CO_3^{2-}	carbonate
OH^{1-}	hydroxide		